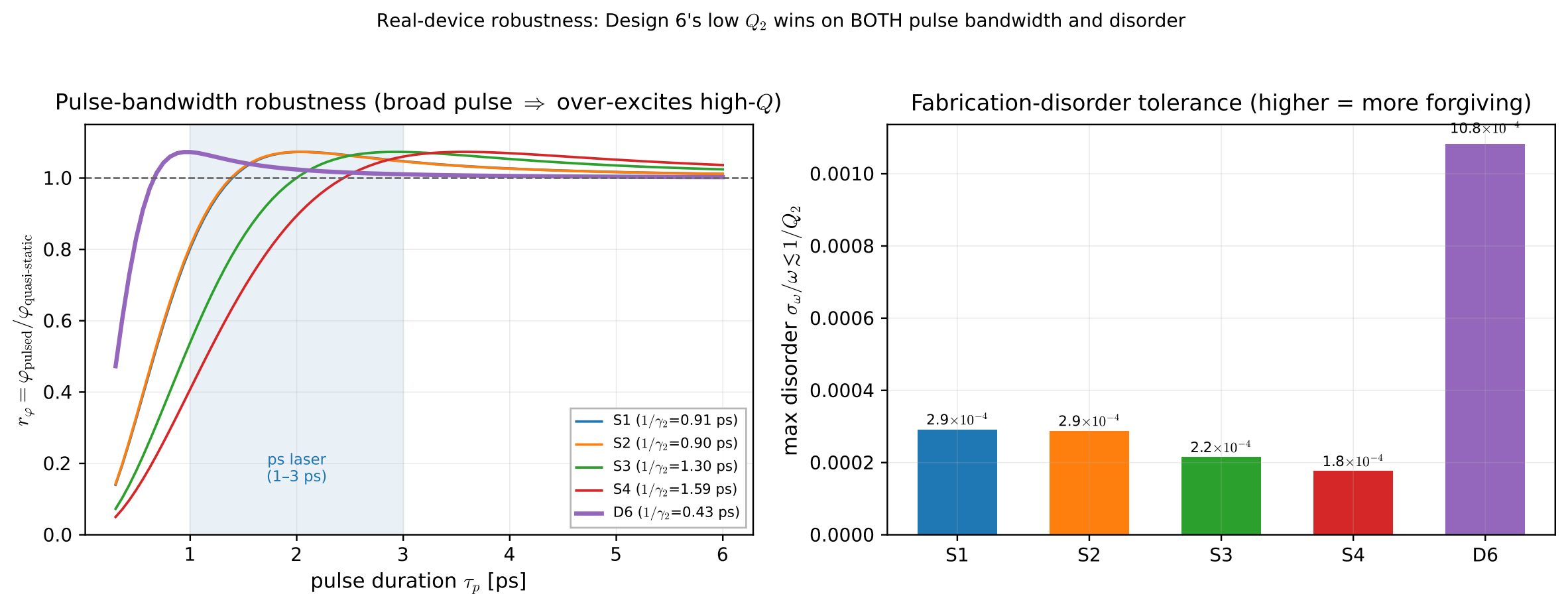
Real-device robustness: Design 6's low Q_2 wins on BOTH pulse bandwidth and disorder
Pulse-bandwidth robustness (broad pulse ⇒ over-excites high-Q)
$r_\varphi = \varphi_{\mathrm{pulsed}}/\varphi_{\mathrm{quasi\text{-}static}}$
pulse duration τ_p [ps]
ps laser
(1–3 ps)
S1 ($1/\gamma_2$=0.91 ps)
S2 ($1/\gamma_2$=0.90 ps)
S3 ($1/\gamma_2$=1.30 ps)
S4 ($1/\gamma_2$=1.59 ps)
D6 ($1/\gamma_2$=0.43 ps)
Fabrication-disorder tolerance (higher = more forgiving)
max disorder $\sigma_\omega/\omega \lesssim 1/Q_2$
2.9×10^{-4}
2.9×10^{-4}
2.2×10^{-4}
1.8×10^{-4}
10.8×10^{-4}
S1
S2
S3
S4
D6